



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to GUHAN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

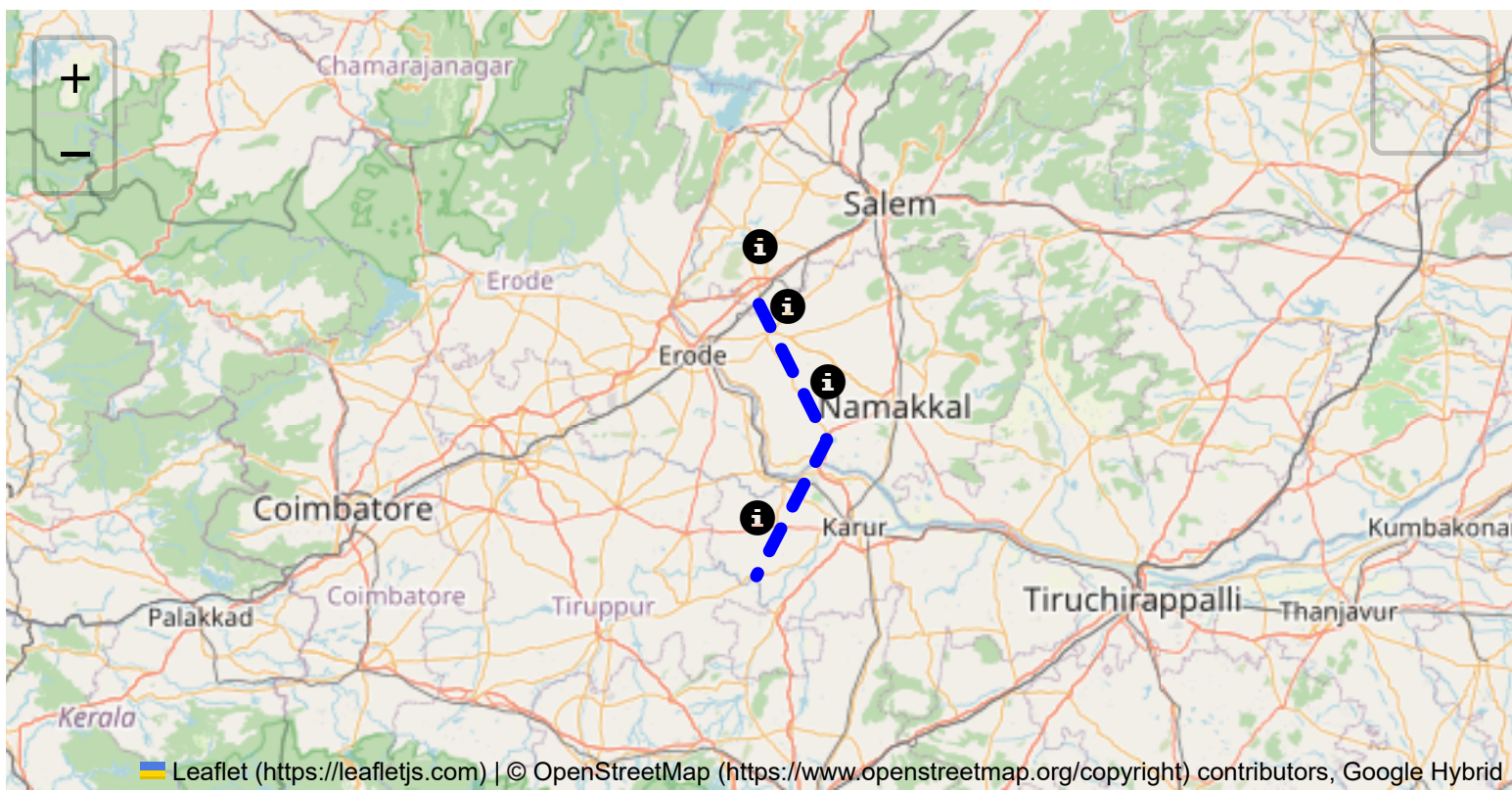
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 91.81 km
Estimated Duration: 1.8 hours
Adjusted Duration (Heavy Vehicle): 2.3 hours
Start: (11.4381, 77.8734)
End: (10.857279, 77.867728)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

The route spans approximately 91.81 kilometers, primarily through rural and semi-urban areas in Tamil Nadu, India. It begins at Sangagiri, passes through Sullipalayam and Paramatti, and ends at Tenilai Venkitapuram. Major roads include the State Highway 86 (SH 86), which is a crucial connector for the region.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

Tamil Nadu generally experiences a tropical climate. It is essential to consider monsoon seasons (June to September and sometimes extending into November) that may lead to heavy rainfall and potential flooding, causing hazardous driving conditions and road closures. The occurrence of thunderstorms is also a possibility during the rainy season.

3. Traffic Patterns and Congestion:

Traffic in this region tends to be moderate, with specific congestion points in and around small towns and near intersections on SH 86. Peak hours typically range from 8:00 AM to 10:00 AM and 5:00 PM to 7:00 PM, where local business and school-related travel increases.

4. Assessment of Road Quality and Infrastructure:

SH 86 is generally in good condition, but some sections may have potholes or narrow bypasses, especially in rural stretches. Road signage might not always be adequate, and lighting can be sparse in less populated areas. Bridges along the route should be checked for load restrictions.

5. Alternative Routes for Emergencies:

In case of road closures or emergencies, consider alternative routes through nearby state highways like SH 79 or SH 95; however, these may increase travel time. Local roads can be used but may not be suitable for heavy vehicles due to narrower passages or lower road quality.

6. Local Regulations Affecting Hazardous Material Transport:

Ensure compliance with Tamil Nadu's regulations on the transport of hazardous materials, including permitted hours, mandatory signage, and adherence to specific routes designated for such transport.

7. Historical Incidents:

There have been occasional incidents involving heavy vehicles due to brake failure, driver fatigue, or loss of control on narrow roads. Any historical spills involving hazardous materials should be investigated further for specific lessons learned.

8. Environmental Considerations and Sensitive Areas:

The route passes near agricultural lands where spills of hazardous materials could have significant implications. Particular care should be taken near water bodies to avoid contamination. Awareness of local wildlife crossings is advised to prevent accidents.

9. Communication Coverage:

Mobile network coverage is generally reliable along SH 86 and in towns. However, intermittent dead zones could occur in remote stretches of rural areas. Pre-planning is advisable for communication needs.

10. Estimated Emergency Response Times:

Emergency response times can vary. Urban centers might see quicker response (approximately 20-30 minutes), while rural areas could experience delays up to an hour or more due to lower infrastructure

support and distance from emergency services.

11. Overall Summary of Risk Assessment:

The route poses moderate safety risks, primarily due to weather conditions, road quality in some sections, and the nature of local traffic patterns. Adequate precautions and adherence to safety regulations will mitigate most potential hazards. Regular updates on weather and road conditions, including any police advisories, will aid in maintaining a safe transport journey. Anticipating environmental and communication challenges will further enhance preparedness on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.96304, 78.06131	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44019, 77.87547	15 KM/Hr
4	Turn	Medium	11.39685, 77.88861	30 KM/Hr
5	Turn	High	11.37862, 77.89488	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16772, 78.01563	30 KM/Hr
8	Turn	Medium	11.15565, 78.02171	30 KM/Hr
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Turn	High	11.14485, 78.02282	15 KM/Hr
11	Turn	Medium	11.06964, 78.00857	30 KM/Hr
12	Turn	Medium	10.94321, 78.00172	30 KM/Hr
13	Turn	Medium	10.93388, 77.99872	30 KM/Hr
14	Turn	Medium	10.90589, 77.97848	30 KM/Hr
15	Turn	Medium	10.89759, 77.96997	30 KM/Hr
16	Turn	High	10.88008, 77.95537	15 KM/Hr
17	Turn	High	10.85769, 77.86796	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium
19	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium
20	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium
21	clinic	Sivabakkiyan Poly Clinic	10.9642254, 78.0549579	30 km/h	Medium
22	hospital	Krishna Medical Centre Hospital	10.9643009, 78.0541215	30 km/h	Medium
23	hospital	vasu medical	10.9638208, 78.0530404	30 km/h	Medium
24	hospital	Meganathan Hospital	10.9658739, 78.052368	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
25	school	st. antonys matric hr. secondary school	10.9641269, 78.0417919	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.96304, 78.06131



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16772, 78.01563



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.15565, 78.02171



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14485, 78.02282



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.06964, 78.00857



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.94321, 78.00172



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.93388, 77.99872



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.90589, 77.97848



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.89759, 77.96997



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.88008, 77.95537



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.85769, 77.86796
